

mielke-geometrodynamics: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image																												
mielke-geometrodynamics_TAB_001	75	—	<pre>\begin{tabular}[t]{ l l } \hline Gauge Field Terminology & Fiber Bundle Terminology \\ \hline spacetime space of phase factors symmetry or gauge group gauge transformations gauge principle classical fields gauge potential gauge field strength & base space bundle space structure group inner automorphisms of the bundle G-equivalence principle cross section of a vector bundle connection 1-form curvature 2-form \\ \hline electromagnetism electromagnetism with monopoles Dirac's monopole quantization & dynamical theory with a connection in a U(1)-bundle connection in a nontrivial U(1)-bundle over \(\mathbb{R}^2 \times \mathrm{S}^2\) classification of U(1)-bundles according to the first Chern class \\ \hline Yang–Mills theory instanton number & dynamical theory with a connection in a SU(2)-bundle classification of SU(2)-bundles over \(\mathrm{S}^4\) according to the second Chern class \\ \hline \end{tabular}</pre>		<table><tr><th>Gauge Field Terminology</th><th>Fiber Bundle Terminology</th></tr><tr><td>spacetime</td><td>base space</td></tr><tr><td>space of phase factors</td><td>bundle space</td></tr><tr><td>symmetry or gauge group</td><td>structure group</td></tr><tr><td>gauge transformations</td><td>inner automorphisms of the bundle</td></tr><tr><td>gauge principle</td><td>G-equivalence principle</td></tr><tr><td>classical fields</td><td>cross section of a vector bundle</td></tr><tr><td>gauge potential</td><td>connection 1-form</td></tr><tr><td>gauge field strength</td><td>curvature 2-form</td></tr><tr><td>electromagnetism</td><td>dynamical theory with a connection in a U(1)-bundle</td></tr><tr><td>electromagnetism with monopoles</td><td>connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$</td></tr><tr><td>Dirac's monopole quantization</td><td>classification of U(1)-bundles according to the first Chern class</td></tr><tr><td>Yang–Mills theory</td><td>dynamical theory with a connection in a SU(2)-bundle</td></tr><tr><td>instanton number</td><td>classification of SU(2)-bundles over S^4 according to the second Chern class</td></tr></table>	Gauge Field Terminology	Fiber Bundle Terminology	spacetime	base space	space of phase factors	bundle space	symmetry or gauge group	structure group	gauge transformations	inner automorphisms of the bundle	gauge principle	G-equivalence principle	classical fields	cross section of a vector bundle	gauge potential	connection 1-form	gauge field strength	curvature 2-form	electromagnetism	dynamical theory with a connection in a U(1)-bundle	electromagnetism with monopoles	connection in a nontrivial U(1)-bundle over $\mathbb{R}^2 \times S^2$	Dirac's monopole quantization	classification of U(1)-bundles according to the first Chern class	Yang–Mills theory	dynamical theory with a connection in a SU(2)-bundle	instanton number	classification of SU(2)-bundles over S^4 according to the second Chern class
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mielke-geometrodynamics_364 TAB_005		—	$\begin{tabular}[t]{l l } \hline := & : \text{Equal by definition} \\ \hline \equiv & : \text{Identically equal} \\ \hline \approx & : \text{Isomorphic} \\ \hline \simeq & : \text{Isomorphic} \\ \hline \sim & : \text{Approximately equal} \\ \hline \sim & : \text{Asymptotic expansion} \\ \hline \times & : \text{Topological product} \\ \hline \otimes & : \text{Direct product} \\ \hline \oplus & : \text{Direct sum} \\ \hline \otimes & : \text{Semidirect product} \\ \hline \mathrm{G}_\circ & : \text{Connected component of the group} \\ \hline \tilde{\mathrm{G}} & : \text{Simply connected covering group} \\ \hline [\mathrm{A}, \mathrm{B}] := \mathrm{AB} - \mathrm{BA} & : \text{Commutator} \\ \hline \{\mathrm{A}, \mathrm{B}\} := \mathrm{AB} + \mathrm{BA} & : \text{Anticommutator} \\ \hline \mathrm{A}^* = \mathrm{C} \mathrm{A} & : \text{Complex-conjugate matrix} \\ \hline \mathrm{A}^+ = \mathrm{A}^{*\mathrm{T}} & : \text{Hermitian adjoint matrix} \\ \hline \pi^{-1}(\mathrm{N}) & : \text{Original domain of N with respect to the mapping } \pi : \mathrm{M} \rightarrow \mathrm{N} \\ \hline \mathrm{T}_{(\mu\nu)} := \frac{1}{2}(\mathrm{T}_{\mu\nu} + \mathrm{T}_{\nu\mu}) & : \text{Symmetrized second-rank tensor} \\ \hline \mathrm{T}_{[\mu\nu]} := \frac{1}{2}(\mathrm{T}_{\mu\nu} - 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